

MSC ELSA 3 — floating debris & pellet drift

As of 4 Jun, 16:00 IST · Prepared by DriftSight · prototype

Summary. 7 satellite passes processed since the sinking on 25 May 2025 (4 usable, 3 clouded). DriftSight has flagged 10 candidate debris patches; 2 are confirmed because they re-appeared where the currents predicted, and 2 look-alikes were rejected before any boat was sent. The pellet forecast puts 88% of the simulated nurdles ashore by this time.

2	2	88%	1
confirmed debris patches	look-alikes rejected	pellets ashore (forecast)	high-priority zones

TRACKED PATCHES

Patch	First seen	Area	Confidence	Status
Patch A	27 May, 11:00 IST	3,300 m ²	99%	confirmed
Patch B	27 May, 11:00 IST	300 m ²	15%	rejected
Patch C	27 May, 11:00 IST	3,200 m ²	96%	confirmed
Patch D	30 May, 11:00 IST	400 m ²	23%	rejected

PELLET LANDFALL OUTLOOK

Thiruvananthapuram 70% (first 27 May) · Kanyakumari 16% (first 29 May) · Kollam 2% (first 27 May) · Ramanathapuram 0% (first 7 Jun). Forecast first arrival at Kanyakumari 29 May, 04:00 IST, consistent with the 30 May beach survey that rated pellet pollution “Very High”.

PRIORITY ACTIONS

- HIGH** — Patch A (at sea) (8.922°N 78.936°E): Boat verification, then boom / skimmer recovery if confirmed. Nearest harbour Pamban.
- MEDIUM** — Patch C (at sea) (8.259°N 78.297°E): Boat verification, then boom / skimmer recovery if confirmed. Nearest harbour Thoothukudi.

MISSIONS

No missions created yet.

DATA & METHOD

Detection: per-pixel classifier on Sentinel-2 bands with FDI/NDVI. Tracking: ensemble particle drift with Bayesian confirmation at each pass. Prototype uses synthetic imagery and a synthetic monsoon circulation; production uses Sentinel-2/Landsat/Sentinel-1, INCOIS/CMEMS currents and ERA5 winds.

Prototype. Detection imagery and ocean fields in this report are simulated; no figure here is a measurement. Production sources: Sentinel-2 / Landsat / Sentinel-1 imagery, INCOIS and CMEMS currents, ERA5 winds.